

What is DevOps?

DevOps is a combination of:

- **Development (Dev)** → people who write the code
- **Operations (Ops)** → people who deploy, manage, and maintain the application/server

Earlier, developers and operations teams worked separately.

This caused:

- delays,
- communication problems,
- deployment failures,
- slow software delivery.

DevOps solves this by making both teams work together continuously.

Simple Definition of DevOps

DevOps is a methodology that combines development and operations teams to automate and speed up software development, testing, deployment, and maintenance.

DevOps Lifecycle

The DevOps process usually follows these stages:

1. **Plan** → Decide requirements
2. **Develop** → Write code
3. **Build** → Convert code into application
4. **Test** → Check for bugs
5. **Release** → Prepare for deployment
6. **Deploy** → Deploy to server/cloud
7. **Operate** → Run and maintain app

Common DevOps Tools

Purpose	Tool
Version Control	Git
CI/CD	Jenkins
Containerization	Docker
Orchestration	Kubernetes
Cloud	AWS, Azure
Configuration	Ansible
Monitoring	Prometheus, Grafana

Advantages of DevOps

1. Faster Software Delivery

Applications are released quickly.

Example:

- New features can be deployed daily instead of monthly.

2. Automation

Testing, deployment, and monitoring are automated.

This reduces manual work.

3. Better Collaboration

Developers and operations teams work together.

Improves communication.

4. Fewer Errors

Automated testing finds bugs early.

5. Continuous Integration & Continuous Deployment (CI/CD)

Code changes are integrated and deployed continuously.

6. Improved Quality

Continuous testing improves software quality.

7. Faster Problem Detection

Monitoring tools quickly identify issues.

8. Scalability

Applications can scale easily using cloud and containers.

What is Agile?

Agile is a software development methodology where work is divided into small parts and developed step by step with continuous feedback.

Instead of building the entire project at once, Agile develops software in small iterations.

Simple Definition of Agile

Agile is a flexible software development approach that focuses on continuous improvement, teamwork, customer feedback, and quick delivery.

Traditional Model vs Agile

Traditional (Waterfall)

Entire project developed at once

Changes are difficult

Customer feedback comes late

Slow delivery

Agile

Developed in small parts

Changes are easy

Continuous feedback

Fast delivery

Agile Process

Agile works in short cycles called **Iterations** or **Sprints**.

Each sprint usually lasts:

- 1 week
- 2 weeks
- or 4 weeks

At the end of every sprint:

- a working feature/product is delivered.

Agile Principles

Agile focuses on:

- Customer satisfaction
- Team collaboration
- Fast delivery
- Continuous improvement
- Flexibility to changes

What is Scrum?

Scrum is one of the most popular frameworks used in Agile.

It helps teams organize and manage work in short sprints.

Simple Definition of Scrum

Scrum is an Agile framework used to manage software development through small time-boxed iterations called sprints.

Scrum Workflow

1. Product Backlog

List of all project requirements/features.

Created and managed by:

- Product Owner (PO)

2. Sprint Planning

Team selects tasks from backlog for the sprint.

3. Sprint

Development work happens here.

Usually:

- 2 weeks
- or 4 weeks

4. Daily Scrum Meeting

Short daily meeting (15 minutes).

Team discusses:

- What was completed yesterday
- What will be done today
- Any blockers/issues

5. Sprint Review

Completed work is demonstrated to stakeholders.

6. Sprint Retrospective

Team discusses:

- what went well,
- what needs improvement.

Roles in Scrum

Mainly 3 important roles:

Role	Responsibility
Product Owner (PO)	Manages requirements
Scrum Master	Manages Scrum process
Development Team	Develops the product

What is Product Owner (PO)?

Product Owner represents:

- customer/business side.

PO decides:

- what features should be built,
- priority of tasks,
- business requirements.

Responsibilities of Product Owner

1. Managing Product Backlog

Creates and prioritizes tasks/features.

2. Understanding Customer Needs

Communicates with stakeholders/customers.

3. Setting Priorities

Decides which feature is most important.

4. Accepting Completed Work

Checks whether work is completed correctly.

Simple Example of Product Owner

Suppose you are building a shopping app.

PO decides:

- login feature first,
- payment feature next,
- then order tracking.

What is Scrum Master?

Scrum Master ensures the Scrum process is followed properly.

They are like:

- facilitator,
- coordinator,
- process manager.

Responsibilities of Scrum Master

1. Conduct Scrum Meetings

Handles:

- daily standup,
- sprint planning,
- retrospectives.

2. Remove Obstacles

Helps team solve issues/blockers.

3. Ensure Scrum Rules Are Followed

Makes sure Agile practices are followed correctly.

4. Protect Team

Prevents unnecessary interruptions.

Simple Example of Scrum Master

If developers cannot access a server:

- Scrum Master coordinates with infrastructure teams and resolves the issue.

Difference Between PO and Scrum Master

Product Owner	Scrum Master
Focuses on product/business	Focuses on process/team
Decides WHAT to build	Decides HOW Scrum runs
Manages backlog	Removes blockers
Customer-facing role	Team-support role

Advantages of Agile

1. Faster Delivery

Software is delivered in small working parts quickly.

2. Easy to Handle Changes

Requirement changes can be added anytime.

3. Better Customer Satisfaction

Continuous customer feedback improves product quality.

4. Improved Team Collaboration

Developers, testers, and business teams work together.

5. Better Quality

Continuous testing reduces bugs.

6. Transparency

Daily meetings provide project visibility.

7. Reduced Risk

Problems are identified early.

Agile + DevOps Relationship

Agile focuses on:

- developing software quickly.

DevOps focuses on:

- deploying and operating software quickly.

Together:

- Agile + DevOps improve complete software delivery.

Simple Real-Time Example

Suppose a company develops a food delivery app.

Agile:

- Team develops features sprint by sprint.
- Example:
 - Sprint 1 → Login
 - Sprint 2 → Order feature
 - Sprint 3 → Payment feature

DevOps:

- Automatically tests and deploys those features to production quickly.

Short Interview Definitions

DevOps

DevOps is a methodology that combines development and operations teams to automate and speed up software delivery.

Agile

Agile is a flexible software development methodology focused on iterative development and customer feedback.

Scrum

Scrum is an Agile framework that manages projects using short iterations called sprints.

Product Owner

Product Owner manages product requirements and prioritizes backlog items.

Scrum Master

Scrum Master ensures Scrum practices are followed and removes blockers for the team.